

## **FLEET RULE OF THE ROAD TEST DEC 14**

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(E)Vol I and the Mariners Handbook (NP100).

**BR454 "A Guide to the Collision Avoidance Rules" 6<sup>th</sup> Edition by Cockcroft and Lameijer was categorised as obsolescent by NCHQ in August 2011. Care should be taken when utilising the 6<sup>th</sup> Edition as the information contained within may not reflect the most recent changes to the regulations.**

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the ROR/training logbook for inspection by FLEET/FOST navigation staff.

In this test, any change in the status of the ship or of the environment will be indicated in the script.

Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

Answers are provided for uniformity of marking. This test is completely scenario based.

**You are OOW in HMS ST ALBANS. You have just sailed from Brest, France on passage to Newcastle. It is the morning watch with fair weather although heavy rain is expected later in the day. Shipping is beginning to increase in volume as you line up for the NE-bound traffic lane of the Dover Straits TSS.**

1. What are you not relieved of with regard to TSS adopted by the Organisation?  
(R.10a)? (1)

2. How does R.10 say you shall use a traffic lane? (R.10) (3)

**As you approach the MPC lightbuoy, you sight the busy port of Calais to starboard. A ferry has just departed from there and is currently at G30, range 4 nm. You are approximately 40 degrees on her PORT bow.**

3. In determining if ROC exists what considerations should be taken into account?  
(R.7) (2)

**From taking the action in your answer to Q.3 you ascertain that a risk of collision does exist.**

4. Who is the give-way vessel and why? (R.15) (2)

5. What action should each vessel take? (R.15, R.16 & R.17) (3)

**You pass the ferry at a safe distance. You notice she has her navigation lights switched on.**

6. When do the Rules state you should exhibit navigation lights? (R.20) (3)

**You hear a VHF transmission from Gris-Nez Traffic that a heavily laden VLCC, constrained by her draft, is following the same NE-bound lane as you are, a few miles behind.**

7. What shapes would you expect this VLCC to display and where? (R.28) (1)

8. How should a vessel of this type navigate? (R.18) (1)

9. What sound signal would this vessel sound, while underway, in restricted visibility and at what interval? (R.35) (1)

**The VLCC's SOA is greater than yours and soon catches up. She is now at G175, on a steady bearing and a reducing range. ROC exists.**

10. Who is the give-way vessel and why? (R.13) (2)

**The situation is resolved. What factors would you consider in deciding a safe speed if you were the OOW in the VLCC and constrained by your draught.**

11. What factors should all vessels consider when determining a safe speed, in any state of visibility? (R.6)

**(6)**

12. When operating within a TSS, a vessel restricted in her ability to manoeuvre, engaged in an operation for the maintenance of safety of navigation, is exempted from what, and to what extent? (R.10k)

**(2)**

**You get to the North Hinder Junction and turn up to the North towards your RAS RV with RFA WAVE KNIGHT. You approach a group of six small vessels each displaying two cones apex together. Some have an additional cone with an apex upwards on one side of the vessel.**

13. What are these vessels likely to be, and what does the additional shape signify? (R.26)

**(1)**

14. What lights would these vessels display by night if making way? (R.26)

**(4)**

15. As you approach, who has the obligation to keep out of the way?

**(1)**

**You rendezvous with RFA WAVE KNIGHT and commence replenishing.**

16. What status could you now claim under the Rules? (R.3)

**(1)**

17. List 3 other activities that a vessel could be engaged in, which would result in this status. (R.3) (3)

18. List 2 types of vessel that could be involved in military operations and claim the same status but are not mentioned in the Rules. (ATP1(E)) (2)

19. What shapes would you display whilst engaged in the replenishment operation? (R.27) (1)

**Quite suddenly a heavy rainstorm engulfs the replenishment group and the visibility reduces to just over 2nm. OOW2 conducts the restricted visibility checklist, switches on the navigational lights and commences restricted visibility sound signals. The radar plot is good and the shipping is light so the CO's of both ships decide to continue with the replenishment.**

20. What is the arc of visibility and range of the masthead light OOW2 has just switched on? (R.21, 22) (2)

21. What sound signal should a vessel Not Under Command make, and at what interval? (R.35) (2)

**You finish the replenishment and continue your passage north but the visibility has now dropped to less than a mile. The plot shows a contact at R45, range 8 miles with a CPA very fine to starboard.**

22. Given that you have detected this contact by radar alone, what actions should you take, and what action should be avoided? (R.19) (2)

23. If you cannot avoid a close quarters situation with this vessel, what action do the rules state you shall take? (R.19) (1)

**You pass the other vessel at a safe distance and the rainstorm passes as quickly as it arrived. Your lookout reports to you an approaching coaster right ahead, in which he can see both masthead lights in a line, and both sidelights.**

24. What situation is developing? (R.14) (1)

25. Who is the give-way vessel and why? (R.14) (2)

**You pass the coaster at a safe distance and the rest of the watch passes quickly and quietly before you hand over and head down below for a well deserved break.**

**Total \_\_\_\_ / 50**

**You are OOW in HMS ST ALBANS. You have just sailed from Brest, France on passage to Newcastle. It is the morning watch with fair weather although heavy rain is expected later in the day. Shipping is beginning to increase in volume as you line up for the NE-bound traffic lane of the Dover Straits TSS.**

1. What are you not relieved of with regard to TSS adopted by the Organisation? (R.10a)? (1)

**(R.10(a)): This Rule applies to traffic separation schemes adopted by the Organization [the IMO] and does not relieve any vessel of her obligation under any rule (1).**

2. How does R.10 say you shall use a traffic lane? (R.10) (3)

**(R.10(b)): A vessel using a traffic separation scheme shall:**

- (i) proceed in the appropriate traffic lane in the general direction of traffic flow for that lane (1);**
- (ii) so far as practicable keep clear of a traffic separation line or separation zone (1);**
- (iii) normally join or leave a traffic lane at the termination of the lane (1/2), but when joining or leaving from either side shall do so at as small an angle to the general direction of traffic flow as practicable(1).**

**As you approach the MPC lightbuoy, you sight the busy port of Calais to starboard. A ferry has just departed from there and is currently at G30, range 4 nm. You are approximately 40 degrees on her PORT bow.**

3. In determining if ROC exists what considerations should be taken into account? (R.7) (2)

**Take a series of compass bearings (1) because...**

**(R.7(d)): In determining if risk of collision exists the following considerations shall be among those taken into account:**

- (i) such risk shall be deemed to exist if the compass bearing of an approaching vessel does not appreciably change;**
- (ii) such risk may sometimes exist even when an appreciable bearing change is evident, particularly when approaching a very large vessel or a tow or when approaching a vessel at close range.**

**From taking the action in your answer to Q.3 you ascertain that a risk of collision does exist.**

4. Who is the give-way vessel and why? (R.15) (2)

**You are (1) because this is a crossing situation under R.15 and the ferry is on your starboard side (1). The presence of the TSS does not give you any special privileges nor does it state a crossing vessel must keep clear of a vessel following a traffic lane.**

5. What action should each vessel take? (R.15, R.16 & R.17) (3)

**Ferry – (R.17(a)(i)): ...keep her course and speed**

**STAL – (R.16): ...take early and substantial action to keep well clear (1) and (R.15): ...if the circumstances of the case admit, avoid crossing ahead of the other vessel (1).**

**You pass the ferry at a safe distance. You notice she has her navigation lights switched on.**

6. When do the Rules state you should exhibit navigation lights? (R.20) (3)

**(R.20(b)): ...from sunset to sunrise (1)**

**(R.20(c)): ..from sunrise to sunset in restricted visibility (1) and may be exhibited in all other circumstances when it is deemed necessary (1)**

**You hear a VHF transmission from Gris-Nez Traffic that a heavily laden VLCC, constrained by her draft, is following the same NE-bound lane as you are, a few miles behind.**

7. What shapes would you expect this VLCC to display and where? (R.28) (1)

**(R.28): A cylinder (1/2), where it can best be seen (1/2).**

8. How should a vessel of this type navigate? (R.18) (1)

**(R.18(d)(ii)): A vessel constrained by her draught shall navigate with particular caution (1/2) having full regard to her special condition (1/2).**

9. What sound signal would this vessel sound, while underway, in restricted visibility and at what interval? (R.35) (1)

**(R.35): ...at intervals of not more than 2 minutes (1/2) three blasts in succession, namely one prolonged blast followed by two short blasts (1/2).**

**The VLCC's SOA is greater than yours and soon catches up. She is now at G175, on a steady bearing and a reducing range. ROC exists.**

10. Who is the give-way vessel and why? (R.13) (2)

**She is (1) because she is overtaking you (1).**

**(R.13(a)): Notwithstanding anything contained in the Rules of part B, sections I and II, any vessel overtaking any other shall keep out of the way of the vessel being overtaken.**

**The situation is resolved. What factors would you consider in deciding a safe speed if you were the OOW in the VLCC and constrained by your draught.**

11. What factors should all vessels consider when determining a safe speed in any state of visibility? (R.6) (6)

**(R.6(a)): By all vessels:**

**(i) the state of visibility;**

- (ii) the traffic density including concentrations of fishing vessels or any other vessels;
  - (iii) the manoeuvrability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions;
  - (iv) at night the presence of background lights such as from shore lights or the backscatter of her own lights;
  - (v) the state of wind, sea and current, and the proximity of navigational hazards;
  - (vi) the draught in relation to the available depth of water.
- [1 mark each]

12. When operating within a TSS, a vessel restricted in her ability to manoeuvre, engaged in an operation for the maintenance of safety of navigation, is exempted from what, and to what extent? (R.10k) (2)

**A vessel restricted in her ability to manoeuvre when engaged in an operation for the maintenance of safety of navigation in a traffic separation scheme is exempted from complying with this rule(1) to the extent necessary to carry out the operation(1).**

**You get to the North Hinder Junction and turn up to the North towards your RAS RV with RFA WAVE KNIGHT. You approach a group of six small vessels each displaying two cones apex together. Some have an additional cone with an apex upwards on one side of the vessel.**

13. What are these vessels likely to be, and what does the additional shape signify? (R.26) (1)

**Vessels engaged in fishing (1/2). When there is outlying gear extending more than 150m horizontally from the vessel, an all round white light or a cone apex upwards in the direction of the gear(1/2).**

14. What lights would these vessels display by night if making way? (R.26) (4)

**(R.26(c)): A vessel engaged in fishing other than trawling shall exhibit:**

**(i) two all round lights in a vertical line, the upper being red and the lower white (1)**

**(ii) when there is outlying gear more than 150m horizontally from the vessel, an all-round white light ...in the direction of the gear (1)**

**(iii) when making way through the water...sidelights (1) and a sternlight (1).**

15. As you approach, who has the obligation to keep out of the way? (R.18) (1)

**You are (1) because:**

**(R.18(a)): a power-driven vessel underway shall keep out of the way of:**

**(iii) a vessel engaged in fishing.**

**You rendezvous with RFA WAVE KNIGHT and commence replenishing.**

16. What status could you now claim under the Rules? (R.3) (1)

**You are restricted in your ability to manoeuvre (RAM) (1)**



17. List 3 other activities that a vessel could be engaged in, which would result in this status. (R.3) (3)

**[Any 3, 1 mark each]**

**(R.3(g)): The term vessels restricted in their ability to manoeuvre shall include but not be limited to:**

- (i) a vessel engaged in laying, servicing or picking up a navigation mark, submarine cable or pipeline;**
- (ii) a vessel engaged in dredging, surveying or underwater operations;**
- (iv) a vessel engaged in the launching or recovery of aircraft;**
- (v) a vessel engaged in mine clearance operations;**
- (vi) a vessel engaged in a towing operation such as severely restricts the towing vessel and her tow in their ability to deviate from their course.**

18. List 2 types of vessel that could be involved in military operations and claim the same status but are not mentioned in the Rules. (ATP1(E)) (2)

**(ATP1(E) Ch 1 Art 2242):**

**Ships engaged in launching or recovery of landing craft utilities (1)**

**Ships engaged in launching or recovery of arrays (1)**

19. What shapes would you display whilst engaged in the replenishment operation? (R.27) (1)

**Ball-diamond-ball (1) because:**

**(R.27(b)): A vessel restricted in her ability to manoeuvre, except a vessel engaged in mine clearance operations, shall exhibit:**

- (ii) three shapes in a vertical line where they can best be seen. The highest and lowest of these shapes shall be balls and the middle one a diamond.**

**Quite suddenly a heavy rainstorm engulfs the replenishment group and the visibility reduces to just over 2nm. OOW2 conducts the restricted visibility checklist, switches on the navigational lights and commences restricted visibility sound signals. The radar plot is good and the shipping is light so the CO's of both ships decide to continue with the replenishment.**

20. What is the arc of visibility and range of the masthead light OOW2 has just switched on? (R.21, 22) (2)

**225 degrees (1) and 6 miles (1).**

21. What sound signal should a vessel Not Under Command make, and at what interval? (R.35) (2)

**(R.35(c)): ..a vessel not under command...shall, instead of the signals prescribed in paragraphs (a) or (b) of this Rule, sound at intervals of not more than 2 minutes (1) three blasts in succession, namely one prolonged followed by two short blasts (1).**

**You finish the replenishment and continue your passage north but the visibility has now dropped to less than a mile. The plot shows a contact at R45, range 8 miles with a CPA very fine to starboard.**

22. Given that you have detected this contact by radar alone, what actions should you take, and what action should be avoided? (R.19) (2)

**(R.19(d)): Every vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists. (1) If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration of course, so far as possible, the following shall be avoided: -**

**(i) an alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken(1/2);**

**(ii) an alteration of course towards a vessel abeam or abaft the beam(1/2).**

23. If you cannot avoid a close quarters situation with this vessel, what action do the rules state you shall take? (R.19) (1)

**(R.19(e)): Except where it has been determined that a risk of collision does not exist, every vessel which hears apparently forward of her beam the fog signal of another vessel, or which cannot avoid a close quarters situation with another vessel forward of her beam, shall reduce her speed to the minimum at which she can be kept on course. She shall if necessary take all way off and in any event navigate with extreme caution until danger of collision is over (1).**

**You pass the other vessel at a safe distance and the rainstorm passes as quickly as it arrived. Your lookout reports to you an approaching coaster dead ahead, in which he can see both masthead lights in a line, and both sidelights.**

24. What situation is developing. (R.14) (1)

**Head-on situation (1)**

25. Who is the give-way vessel and why? (R.14) (2)

**You both are (1) because:**

**(R.14(a)): When two power-driven vessels are meeting on reciprocal or nearly reciprocal courses so as to involve risk of collision each shall alter her course to starboard (1) so that each shall pass on the port side of the other.**

**You pass the coaster at a safe distance and the rest of the watch passes quickly and quietly before you hand over and head down below for a well deserved break.**

**Total \_\_\_\_ / 50**